What TRDP is building

A visual dataset for testing whether visible body context changes the representation of observed index-finger movement.

One source pair: static frame A and movement frame B. The index opens away from the middle finger; the camera and scene stay fixed. A static trial presents A then A. A movement trial presents A then B.

One family: one source + one desk + one random seed + one shared white balance. It contains A/B images in three contexts: full body, upper limb and detached hand.

20
approved V7 source pairs

160
shared desk generations

480
appearance families

2,880
saved condition PNGs

A / Build approved sources Pages 2-9	Original reference pair -> guided scene -> hand mask and clean desk plate -> shared hand repair -> geometric movement -> bounded movement repair -> shared clothing correction -> V7 human approval. These historical operations are already baked into the frozen source pixels.
B / Run current expansion Pages 10-20	Load frozen V7 pairs and masks -> propose desk texture -> blend only the editable desk, identically in A/B -> apply shared white balance -> export three contexts -> verify saved files -> human review. The clean repository implements this stage with a corrected mask. Accepted-batch examples retain the original mask version (page 13).
History and next steps Pages 21-25	Concise ledger of earlier methods, existing failure images, current validation limits and the next scientific evaluation steps.

All photographs here are real saved project artifacts. Crops and overlays explain them; PDF previews are reduced copies. The local PNGs retain their original lossless pixels. Image acceptance does not establish anatomical or physiological validity.

1 / Start with the original reference pair

Worked example: male reference 12. These grayscale reference photographs establish scene layout and the intended index opening.

INPUT A / Original static reference



The hand rests palm-down; fingers are close together.

INPUT B / Original movement reference



The index opens away from the middle finger.

How the two references are used

The static reference supplies the Canny outline for scene generation. The reference pair supplies the desired image-plane opening. The workflow creates one appearance parent, then moves its index; generating A and B independently would change unrelated scene details.

Saved evidence: TRDP1_corrections/controlnet-input/{normal12,open12}.png. Original reference size: 1254 x 1254. Reference-12 movement target: 18.3704 degrees of image-plane rotation.

2 / Generate the photorealistic appearance parent

Read left to right: original layout -> Canny edges -> one new static scene. This is text-to-image generation guided by edges.

INPUT / Static reference



Reference resized with Lanczos to 1024 square.

GUIDE / Canny outline



Canny thresholds 30 / 75; repeated to three channels.

OUTPUT / Static parent



One generated scene, enlarged to 2048 with Lanczos.

Recorded source-generation settings

RealVisXL V4.0 + Xinsir Canny SDXL + fixed fp16 VAE. DPM++ multistep / Karras; 30 effective steps; CFG 4; control scale 0.9; control end 0.8; seed 9. Render 1024 x 1024, fp16, CPU offload. Both text encoders receive the same prompt. This source recipe differs from the later desk-inpaint recipe.

Positive prompt: First-person photograph of a seated adult at an empty wooden desk. One left hand flat, palm down, four distinct straight fingers close together, relaxed thumb. Entire hand and forearm visible. Natural knuckles, realistic skin and short nails. Neutral long sleeves and trousers, legs below the desk. Soft diffuse daylight, fixed camera, coherent perspective.

Saved evidence: appearance_heldout/p12_seed9_detailed/{control.png,appearance_1024.png,static_full.png,result.json}. Reference 12 was held out from the initial four-reference prompt screen.

3 / Separate hand pixels from the desk

A hand matte tells the movement builder which pixels belong to the hand. ObjectClear supplies desk texture for places the hand vacates.

INPUT / Hand area



Native crop: x 293-910, y 165-782 (617 square).

HAND MATTE / Foreground



White owns hand pixels; black belongs to the background.

OUTPUT / Empty desk plate



Saved ObjectClear result; a texture donor for movement.

Method and recorded parameters

Author SAM-HQ segmentation plus reviewed gap/contour corrections defined the matte. The 1024-square matte is resized into the native 617-square crop. ObjectClear removed the full hand contour at CFG 2.5, seed 10031, 20 evaluations; its saved raw output is 512 square. The plate is blended locally, with contact/shadow handling, rather than replacing the whole scene.

Two different meanings: this white-hand matte selects foreground to move. The later white-desk mask (page 11) selects background that may be regenerated. Their colors describe different operations.

Saved evidence: regeneration_v5/frozen_v4_inputs/p12_seed9_detailed/hand_alpha.png and fresh_pilot_v2/results/objectclear_plate_e66f77618140dbef/p12_seed9_detailed_movement_10031/.

4 / Repair the shared hand appearance

V5 first corrected nails and hand texture once in the static source, before transporting those pixels into movement.

BEFORE / Generated hand



The same 617-square native hand area.

PERMISSION / Repair mask



White permits local repair; black preserves the source.

AFTER / Shared hand repair



Selected repair is reused in the static and movement construction.

Recorded local-inpaint settings

Same pinned SDXL / Canny / VAE stack and DPM++ Karras. Native crop 617 square -> render 1024 square -> Lanczos back to native. 40 requested steps x strength 0.35 = 14 effective steps; CFG 3.5; control scale 0.8; end 0.85; seed 9. Blend by the saved repair weight. Changed pixels: 106,210; changes outside mask: 0.

Positive prompt: Close-up photograph of a real adult left hand resting flat palm-down on a wooden desk, five naturally proportioned fingers, realistic short unpainted fingernails, natural skin, subtle veins and knuckle creases, soft diffuse daylight, coherent contact shadows, consistent skin tone, sharp photographic detail.

Saved evidence: regeneration_v5/shared_hand_batch/p12_seed9_detailed_source/{comparison.png,composite.png,result.json}. References 3 and 5 used the medium repair-screen selection; this is reference 12.

5 / Construct the index movement geometrically

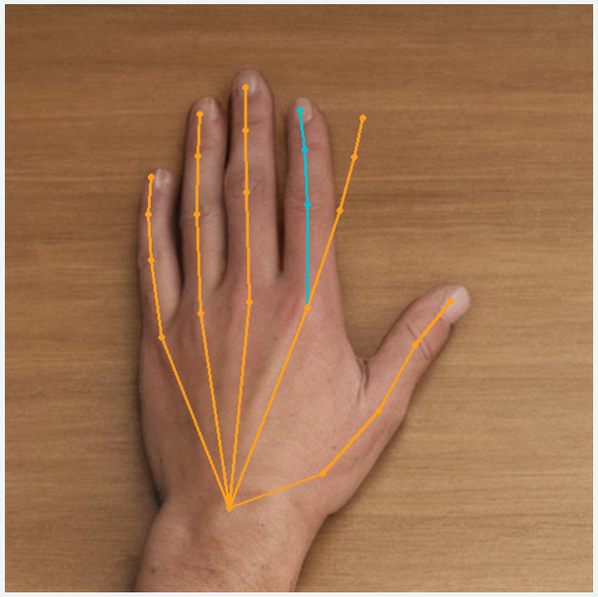
The corrected source hand is transported through the audited geometry. The old ObjectClear desk plate is reused; no new removal call is claimed.

INPUT / Repaired source



Start from the same hand texture and appearance.

GEOMETRY / Fixed anchors



Cyan: source points. Orange: target index; other joints stay fixed.

OUTPUT / Movement initializer



A geometric initializer; the finger root still needs local repair.

How movement is made

Rotate the upper index rigidly around its base (landmark 5) by 18.3704 degrees. A harmonic piecewise-affine mesh joins the root smoothly to the stationary palm. The 21 initialization handles map exactly; only index joints 6-8 move. Reuse the desk plate where the finger vacates, transport the shared hand repair through the same warp and lock stationary hand pixels.

Limit: this is an image-plane construction with approximate contact/shadow changes. Exact initializer coordinates do not certify the final repaired anatomy or a true 3D hand rotation.

Saved evidence: regeneration_v5/prepare_movement_batch.py and frozen_v4_inputs/p12_seed9_detailed/case.json. Anchor overlay is drawn from that saved case, not redetected for this guide.

6 / Repair the movement edge and finger web

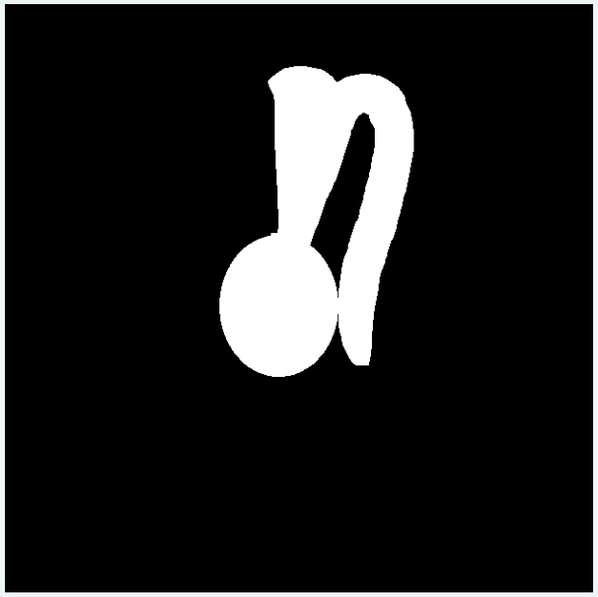
V5 used a bounded local inpaint after geometry, then restored all pixels outside the permitted support.

BEFORE / Geometry initializer



Geometry and shared hand texture already present.

PERMISSION / Local support



Finger boundary, vacated air and root; stationary digits locked.

AFTER / Selected movement



Selected seed-123 repair, returned to the native 617-square crop.

Recorded local-inpaint settings

40 requested steps x strength 0.45 = 18 effective steps; CFG 3.5; control scale 0.4; end 0.85; seed 123; render 1024 square; DPM++ Karras. Canny 50 / 110, with editable root edges removed from control. The mask includes a narrow boundary band and root ellipse; the saved weight fades inward. Changed pixels: 38,778; changes outside mask: 0.

Positive prompt: Close-up photograph of one adult left hand resting palm-down on a wooden desk, index finger slightly abducted from the middle finger. Natural continuous finger web and knuckles, five anatomically correct digits, short realistic nails, fine skin texture. The hand touches the desk, soft contact shadows, matching diffuse daylight, consistent skin tone and wood grain.

Saved evidence: regeneration_v5/movement_batch_v2/p12_seed9_detailed_movement/{comparison.png,composite.png,result.json}. The first seed-9 cohort was superseded after rough-knuckle feedback.

7 / Make the torso and sleeves one coherent garment

V6 corrected clothing jointly, then composited only the lower scene. The repaired body is copied identically into A and B.

BEFORE / V5 clothing



Torso texture and seams did not match the sleeves.

PROPOSAL / Joint garment edit



Built-in image editor: torso and both sleeves corrected together.

AFTER / Protected composite



Use the edited lower scene; restore the original upper desk and hand.

Recorded method and parameters

Prompt asks for one light heather-grey cotton sweatshirt with plausible folds, matching sleeves and no mismatched fabric layers, torso seams or thick folded band. Editor output: 1254 square; enlarge with Lanczos only for the clothing/lower-scene composite. Reference-12 desk edge: y = 0.001 x + 1255.2. Preserve 5 px below it, then use a 24 px transition. Hand-crop changes: 0; changes outside lower-scene support: 0.

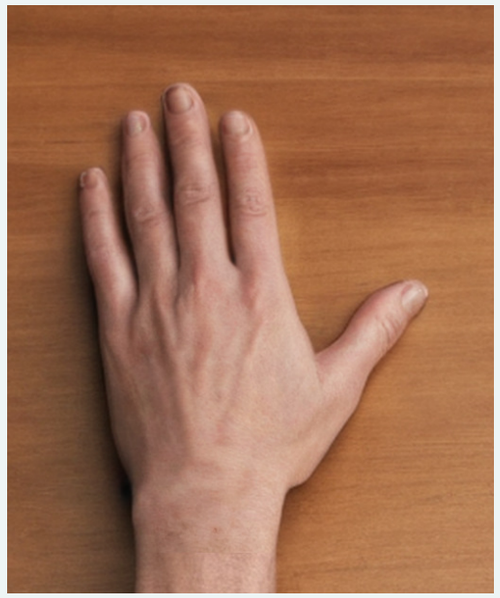
Reproducibility limit: the built-in editor did not expose a seed or pinned model revision. Its selected result is reproducible as stored frozen pixels. Re-running the prompt is not an exact reconstruction.

Saved evidence: regeneration_v6/joint_garment_edits/p12_seed9_detailed/{raw.png,composite.png,weight.png,result.json}. The composite and frozen V7 static have identical pixels.

8 / Resolve the last V7 source issue and approve

Reference 2 needed an additional correction because its movement index looked bent. This page changes examples explicitly: it is reference 2.

REFERENCE 2 / Static A



Original index appearance to preserve.

BEFORE / V6 movement



User flagged the changed finger shape.

AFTER / Accepted V7 movement



Original shaft rigidly rotated; local rim/root restored.

Reference-2 correction and final approval

Rotate the original index by 18.8889 degrees; source cutoff y 270, 16 px transition. Register the local editor repair (maximum 28 px shift), color-match skin/desk residuals (Gaussian sigma 20 px, offset capped at 30) and blend the root over 12 px. Preserve 6,088 original rigid-shaft pixels exactly. Editor seed/model are unavailable. Detector discrepancy: 0.427 degrees; diagnostic only.

V7 freeze: user accepted all 20 source pairs, including reference 18. All 60 static condition images and the other 19 families' 114 condition images stayed byte-identical. Historical pre-approval result files still say pending; final acceptance supersedes that status.

Saved evidence: regeneration_v7/root_edit_restored_v2/registered/result.json, USER_ACCEPTANCE_V7.json and FINAL_VERIFICATION_V7.json. Reference 12 has no additional V7 hand edit.

9 / Freeze the inputs used by the current generator

From here onward the clean repository runs the workflow. It consumes reviewed V7 pixels and stored preparation inputs.

V7 STATIC A



Approved static source, RGB 2048.

V7 MOVEMENT B



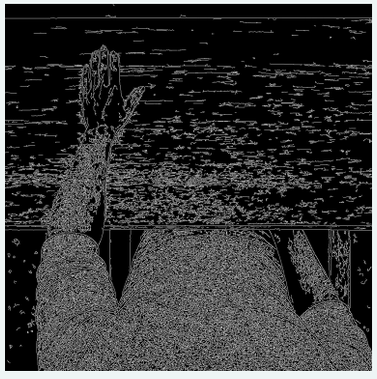
Approved movement source, RGB 2048.

INITIAL IMAGE / 1024



Static A resized once with Lanczos.

CANNY CONTROL / 1024



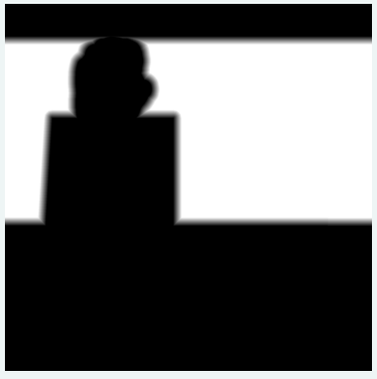
Canny 30/75; three RGB channels.

BINARY DESK MASK / 2048



White: edit desk. Black: preserve.

BLENDING WEIGHT / 2048



Black: preserve. Gray: blend. White: replace.

Frozen bundle: 20 sources x (six context PNGs + five input images) = 220 files. The fifth input is mask_review.jpg, a colored desk overlay for checking. data/frozen_sources.json binds every file hash and its source-specific crop geometry. Checks reject altered or missing files.

10 / Understand the desk mask in your screenshot

A mask is an editing permission map. It is not a photograph, a target output or something participants see.

INPUT / Approved source



The source hand, contact and body must survive.

OVERLAY / Editable desk



Cyan marks where desk editing is allowed.

MASK / Permission map



White permits desk edits; black protects everything else.

Why the hand shape is black

The mask protects the union of the static hand, target hand and all pixels that differ between A and B. A 36 px guard keeps editing away from hand contact. A generous arm corridor protects the forearm. The top strip (y < 180) and everything below 18 px above the fitted desk edge are also protected. The source hand matte is correctly placed in native crop coordinates before these guards are built.

It does not paint the desk white: white tells SDXL where it may propose new pixels. Black areas are copied from the approved source during the final composition. This is the corrected current mask; the accepted archive used the earlier version shown on page 13. The two frames receive exactly the same edited desk.

Saved evidence: data/inputs/p12_seed9_detailed/{desk_mask.png,mask_review.jpg}. Preparation: dataset_production_2026_10_11/prepare_production.py; frozen guard 36 px, inward feather 48 px.

11 / Use a soft weight to hide the desk boundary

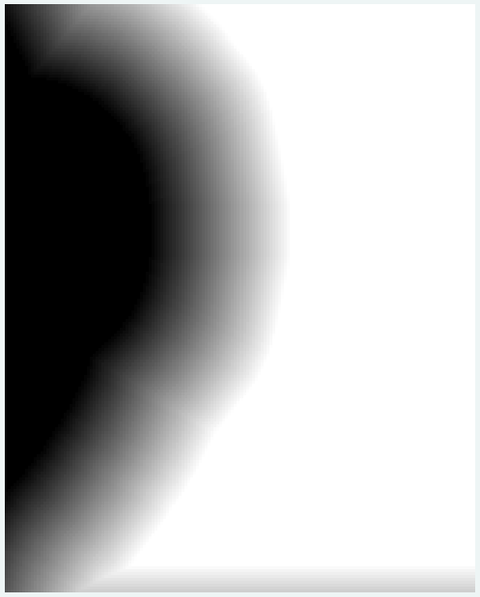
The binary permission mask and grayscale blending weight are different files. The weight prevents an abrupt pasted edge.

BINARY MASK / Edge close-up



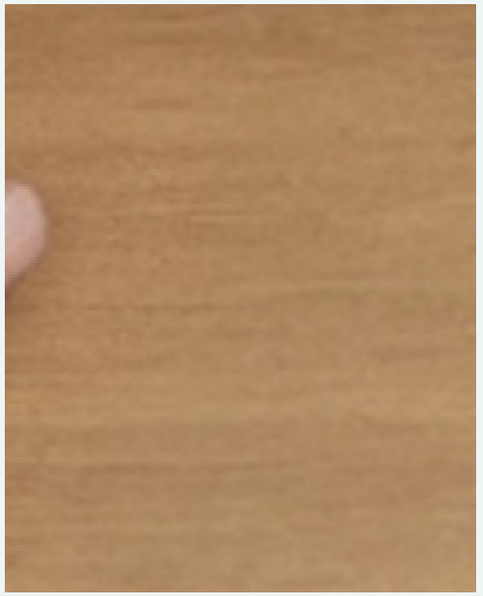
Only black 0 and white 255; passed to inpainting.

SOFT WEIGHT / Same close-up



A 48 px inward fade from black through gray to white.

DEMONSTRATION / Recomposition



Old proposal recomposed with the corrected guard; not an accepted export.

How a weight becomes a pixel

Weight 0 means 100% approved source. Weight 255 means 100% color-matched desk proposal. A middle value mixes the two. For example, weight 128 gives about half source and half proposal. The weight is round(255 x min(distance inside editable desk / 48, 1)). The binary desk mask is white wherever the stored weight is greater than zero.

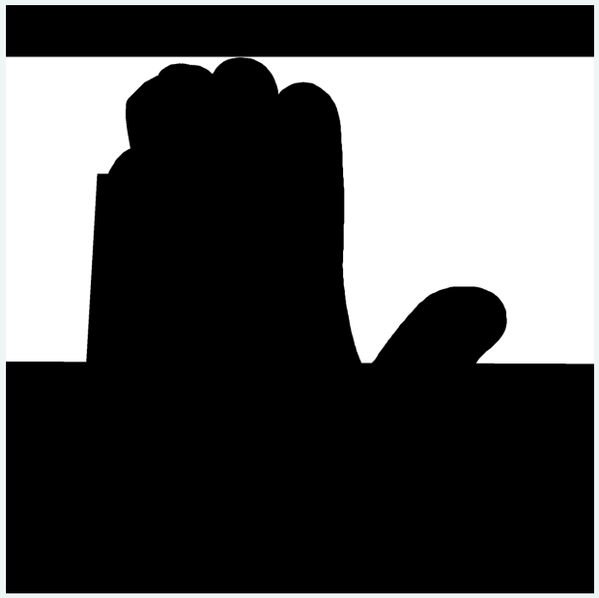
Exact protection: final composition explicitly restores all weight-zero pixels. White-balance changes are applied later, identically to both frames; verification compares protected pixels after that same transform.

Saved evidence: data/inputs/p12_seed9_detailed/desk_weight.png and scripts/dataset.py: shared_desk / audit_pair. Zoom uses real saved mask pixels; it is enlarged only for explanation.

Mask versions / preserve the approved batch

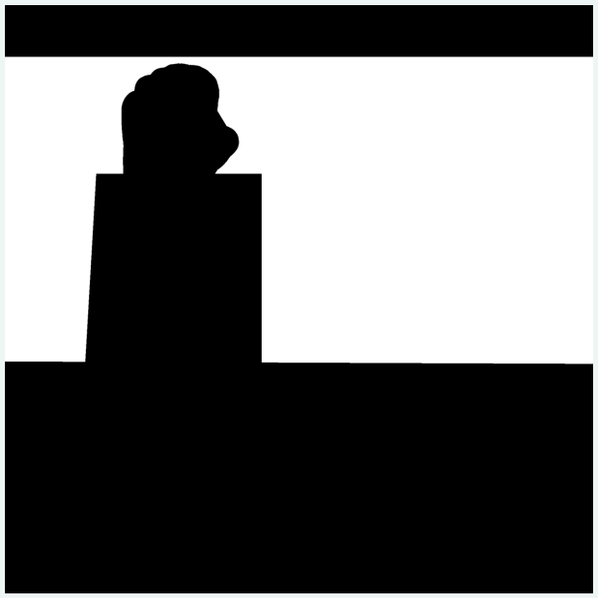
Your screenshot is exactly the original reference-1 binary mask. Its large hand silhouette comes from the earlier crop-coordinate mistake.

BEFORE / Original reference-1 mask



Matches your screenshot pixel-for-pixel. Used for the accepted batch.

AFTER / Corrected current mask



The hand matte is resized to its native crop, then placed in the full scene.

CHECK / Corrected photo overlay



Cyan editable desk now follows the actual hand/contact protection.

Two versions, two honest validation scopes

The original 480 accepted families are preserved unchanged. Their original desk weights and 160 diffusion-parent records are stored locally under data/accepted/_provenance/. The corrected masks in data/inputs/ are for future clean-code generation. All 480 accepted families have pixels differing inside the corrected guard. Original hashes and original-mask checks pass; passing them does not imply passing the new guard.

Decision: preserve the approved archive and document the corrected setup. A new corrected-mask run needs fresh GPU sampling and visual approval. The remaining reference-12 desk example uses the original mask so every shown accepted output can be reproduced from its saved proposal.

Saved evidence: Original weights are hash-bound to each family via its diffusion-parent result. Current frozen_sources.json records both corrected input hashes and previous input hashes.

12 / Generate one new desk proposal

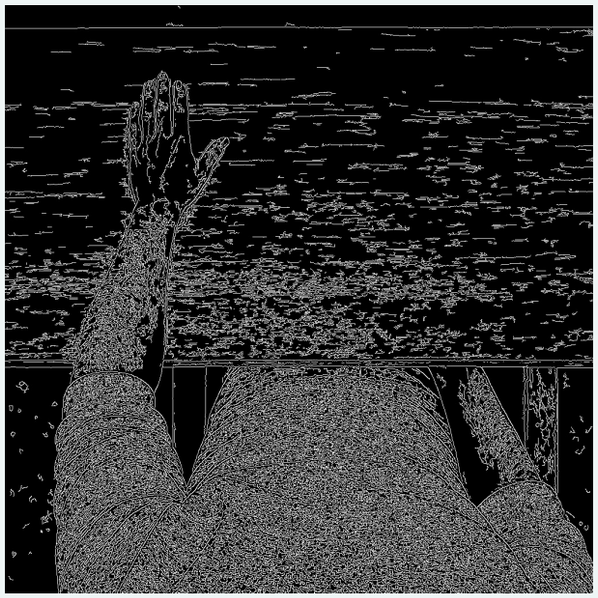
Accepted-batch example: reference 12, birch, seed 9, original mask. A fresh corrected-mask proposal has not been sampled.

INPUT / V7 static A



Initial image resized to 1024; binary mask resized with nearest neighbor.

GUIDE / Stored Canny



Stored 1024-square edges constrain desk perspective.

PROPOSAL / Raw SDXL result



Raw 1024 output; the whole image is not accepted as a stimulus.

Recorded accepted-run inpaint settings / same recipe in current code

RealVisXL V4.0 + Xinsir Canny SDXL + fixed fp16 VAE; DPM++ multistep / Karras. Render 1024 square; 40 requested steps x strength 0.4 = 16 effective steps. CFG 3.5; control scale 0.65; control end 0.9; seed 9. fp16; model CPU offload, VAE tiling/slicing, two CPU threads. Raw proposal enlarged with Lanczos to 2048.

Positive prompt, birch: Photograph of a clean warm natural birch wood with delicate irregular grain desktop. Subtle realistic wood texture, even soft daylight, original camera perspective. Keep the desk flat and empty, preserve its straight edges and existing contact shadows. The seated adult, hand, forearm and neutral sweatshirt stay unchanged. Natural unretouched photograph.

Saved evidence: config/recipe.json and original diffusion_parents/r12_birch_s9/raw_1024.png. Raw hand/body pixels may change; they are discarded by the protected compositor on the next page.

13 / Blend the desk and share it between A and B

Only permitted desk pixels enter the final images. The approved movement and stationary body are kept from the frozen source pair.

INPUT / Raw proposal



Lanczos enlargement provides proposed desk pixels.

OUTPUT A / Shared desk



Native source A + the color-matched, weighted desk.

OUTPUT B / Same shared desk



Native source B + exactly the same editable desk pixels.

Composition, in order

1. Compare source and proposal median RGB over the desk core (weight >= 250). Clip the correction to +/-24 per channel; actual example offset: [-4.0, 0.0, -2.0].
2. Blend the corrected proposal into A using desk_weight / 255; explicitly restore weight-zero pixels.
3. Start B from original movement B and copy A's editable desk pixels into B. Reject any mask/action overlap.

Verified historical example: this guide recomputes the reference-12 birch/seed-9 composite from the saved raw proposal and original mask. It matches the original saved shared-static pixels exactly. The warm export also matches the original saved PNG pixels.

Saved evidence: scripts/dataset.py: shared_desk. Source action pixels: 49,294; original editable desk pixels: 1,215,027; overlap: 0. The raw proposal is not one of the six condition images.

14 / Derive three shared white balances

Neutral, warm and cool are deterministic linear-RGB transforms of the same composed pair. No additional diffusion run is required.

NEUTRAL / static A



Linear-RGB gains [1.0, 1.0, 1.0]

WARM / static A



Linear-RGB gains [1.035, 1.0, 0.965]

COOL / static A



Linear-RGB gains [0.965, 1.0, 1.035]

NEUTRAL / movement B



Linear-RGB gains [1.0, 1.0, 1.0]

WARM / movement B



Linear-RGB gains [1.035, 1.0, 0.965]

COOL / movement B



Linear-RGB gains [0.965, 1.0, 1.035]

Operation: convert sRGB to linear RGB, multiply red/green/blue by the listed gains, clamp, convert back to sRGB and round to bytes. The same gains apply to A and B, including the protected hand/body. Geometry and shadow positions stay fixed.

Previews crop out part of the lower frame to show the desk; exports remain 2048 square. These are white-balance variants, not independent lighting simulations or new source identities.

15 / Export the six condition images

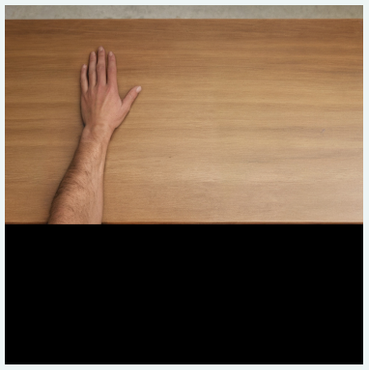
One accepted example family: r12_birch_s9_warm. Every file remains a 2048 x 2048 RGB PNG with its original coordinate system.

FULL BODY / static



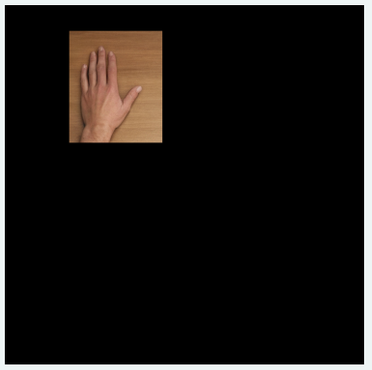
A to A trial uses static twice.

UPPER LIMB / static



A to A trial uses static twice.

DETACHED HAND / static



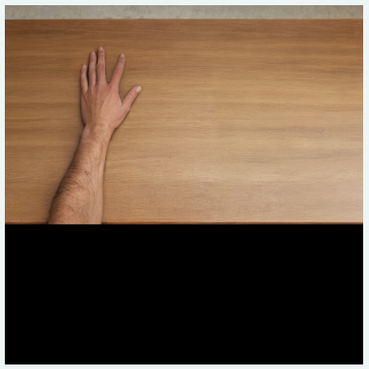
A to A trial uses static twice.

FULL BODY / movement



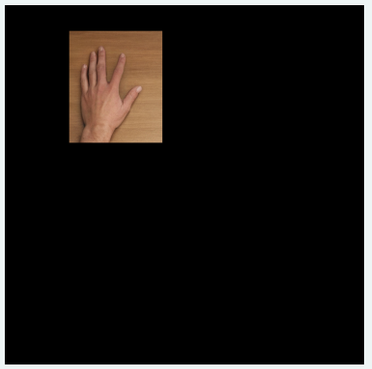
A to B trial uses static then movement.

UPPER LIMB / movement



A to B trial uses static then movement.

DETACHED HAND / movement



A to B trial uses static then movement.

Full body: keep all pixels. **Upper limb:** keep rows y < 1249 for reference 12; black below. **Detached hand:** keep rectangle [367, 147, 898, 784); black elsewhere. Bounds are source-specific, stored in frozen_sources.json. Visible pixels are exactly the same across contexts.

The detached-hand view keeps the hand ROI and its small desk surround; it is a rectangular context manipulation, not a skin-only silhouette. No resizing or repositioning is applied to the saved conditions.

16 / Expand to 480 families without changing motion

The factorial grid varies desk appearance, generation seed and white balance. Three contexts are exported from every resulting pair.

20 approved source pairs	One frozen A/B pair per source group. Source-generation seed is 9 for 18 sources and 123 for references 10 and 18. These source seeds are separate from the expansion seeds below.
4 desks x 2 seeds	Beech: fine straight grain. Oak: subtle open grain. Maple: fine flowing grain. Birch: delicate irregular grain. Seeds 9 and 123 are used for each source/desk. 20 x 4 x 2 = 160 SDXL desk parents.
3 shared white balances	Each desk parent becomes neutral, warm and cool. 160 x 3 = 480 appearance families. Each source therefore contributes 24 families; each family contributes six condition PNGs.
3 contexts x 2 frames	Full body, upper limb and detached hand, each with static and movement. 480 x 3 x 2 = 2,880 condition PNGs. Six conditions are paired views of one family, not independent samples.
Stable naming and grouping	r12_birch_s9_warm = reference 12, birch desk, expansion seed 9, warm white balance. source_id p12_seed9_detailed identifies its frozen parent. source_lineage_group reference_12 keeps the source variants grouped.
Known accepted issue	Reference 17 has a flagged clothing issue. Keep its flag in metadata; include/exclude it explicitly in the analysis protocol. Removing its 24 families leaves 456 families / 2,736 PNGs.
What the number does not mean	480 families do not mean 480 people, poses or independent source photographs. Audit common upstream origins before splitting, and merge related source groups where needed.

17 / Verify files, then review their appearance

Automated invariants catch corruption and unintended edits. Human review decides whether textures, anatomy and contact look acceptable.

1. Frozen inputs	Check hashes of all 220 source/input files. Decode them; check native dimensions, formats and modes. Ensure binary mask equals weight > 0, desk does not overlap action, and source contexts retain the correct pixels.
2. Saved output invariants	Reopen each saved PNG, verify RGB / 2048 square and its recorded hash. Protected pixels equal the frozen source after the same white balance. A/B differences stay inside original action support and remain nonempty. Contexts retain full-body pixels exactly; hidden pixels are black.
3. Ten canaries first	Generate only the ten IDs in config/recipe.json. Visually inspect all six PNGs per family: natural table texture, smooth transition, preserved hand/contact, preserved body/clothing, correct pair action and natural lighting. Record every check and an explicit reviewer decision.
4. Gate the full run	Only ten approved canaries unlock full generation. Afterward review the remaining families. Numeric success and historical approval never silently approve fresh output. Review records bind to the generated files and run provenance.
5. Keep the accepted archive	data/accepted// contains six original PNGs plus audit.json. validate_dataset.py checks approval and hashes, then protected/action/context pixels using the original hash-bound weights. It separately counts deviations from the corrected guard and writes local validation.json.
6. Interpret old metadata	The original audit.json can retain a numeric-pass / review-pending status from generation time. docs/accepted-dataset.json and approval-receipt.json carry later user acceptance. Both are preserved; the archive validator checks the final approval record.
Limits of these checks	Hash and pixel checks establish integrity and constrained processing. They do not certify photorealism, independent biological identities, exact 3D kinematics or physiological replication. Fresh GPU sampling of the moved clean code remains to be checked.

18 / Navigate and run the clean repository

Code is grouped by purpose. The README contains the complete commented directory tree, every command and the full local archive layout.

Repository map

```
config/recipe.json
scripts/setup.sh   - review / CUDA environment
scripts/models.py   - pinned model cache
scripts/run.py     - cooperative launcher
scripts/generate.py - CUDA desk worker
scripts/dataset.py  - checks, export, review
scripts/validate_dataset.py - original archive
tests/test_dataset.py - CPU regressions
data/frozen_sources.json - input hashes
docs/ - guide, validation, approval metadata
README.md, requirements.txt
.gitignore, .gitattributes
```

Local review commands

```
./scripts/setup.sh review
source .venv/bin/activate
python -m unittest discover -s tests -v
python scripts/dataset.py check
python scripts/validate_dataset.py
```

CUDA generation entry points

```
./scripts/setup.sh cuda
python scripts/models.py --download
python scripts/generate.py canary --preflight
python scripts/run.py canary
# Review and approve before full generation.
python scripts/run.py full --cooperative
```

Local data layout and storage

data/sources///{static,movement}.png stores the 120 V7 images. data/inputs// stores the five frozen preparation inputs. data/accepted///{static,movement}.png stores all 2,880 accepted images, plus one audit.json per family. The original weights and parent records are in data/accepted/_provenance/. The unchanged 480-page review stays in docs/480-family-review.pdf. All large data, environments, model caches and new output stay outside Git tracking.

CUDA target: Linux / NVIDIA, at least 12,000 MiB free VRAM and 16 GiB free output space; pinned PyTorch CUDA 12.8 wheels. Mac runs review and integrity checks. Cloning GitHub alone does not download local data; copy the four local paths listed in the README.

The public repository contains the corrected V7 expansion system; accepted originals remain historical. Earlier creation scripts and unused experimental models remain in the original project folders, with their method history summarized next.

Earlier methods / models, controls and refinement

Grouped ledger of tried alternatives. A rejected setting or memory failure does not establish that the method can never work.

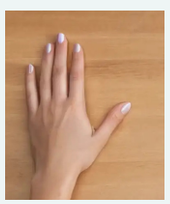
Unguided scene generation	DreamShaper SD1.5 at 768 and DreamShaperXL v2 Turbo at 1024 were compared. Unguided anatomy and composition were unreliable; guided SDXL became the base.
Control types	Canny, MiDaS depth, OpenPose, SegFormer-B5 and PidiNet soft edges were tried. First-person pose and segmentation were unreliable; soft edges weakened boundaries. Canny was retained. Canny + depth weakened edges and took about 33% longer in the earlier comparison.
Checkpoint comparison	SDXL Base, JuggernautXL v9, NightVisionXL, RealVisXL V4, Pony Realism v2.3, epiCRealism SD1.5, Realistic Vision 5.1/6 and SD3.5 Medium were screened. RealVis best fit the chosen photorealism tradeoff. FLUX.1 dev ran out of memory on 16 GB; its image quality was not evaluated.
Parameters and schedulers	CFG 1-9, Canny scale 0.3-1.1, control end 0.6-1 and 10-60 steps; Euler, DPM++ Karras, UniPC and DDIM were compared. The older CFG 4 / control 0.9 / end 0.8 / 30-step setting became the source recipe. Final desk inpaint uses CFG 3.5 / control 0.65 / end 0.9 / 40 nominal steps at strength 0.4.
Appearance / prompt screens	Original/scene prompts, blurred/high-threshold Canny, img2img, SDXL Base and early control stop: high thresholds lost boundaries; early stop bent contours. DPM++ helped. Baseline/detailed/detailed-negative/dual-encoder screens had 14/16, 16/16, 16/16 and 14/16 numerical passes; that is not a general visual-yield estimate.
Upscaling and refinement	Real-ESRGAN x2/x4, BSRGAN, Swin2SR x4, SD x4 Upscaler, ControlNet Tile x2 and SDXL hires fix were compared. Hires fix was used earlier. Final expansion uses Lanczos to preserve approved anatomy. Uncontrolled, controlled and 80-nominal-step refinement gave little benefit; Tile/Swin2SR could degrade structure.
Original movement inpaint	Rectangular/wider masks, whole-hand crops, disabled registration, dedicated inpaint (baseline, stronger, lower control, combined), no ControlNet, erase-redraw, lower control and DPM++ were tried, including holdouts. Failures included protected-pixel changes, ghosts, adjacent-digit changes and translucency. Collar-tone and Poisson blending did not repair bad raw anatomy.

Earlier methods / learned editors and cutouts

The two failure examples below are saved experimental outputs. No new failure images were generated for this report.

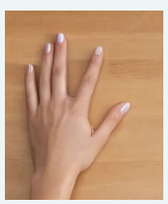
FoundHand / 12 trials	Reposing changed bend, nails and shading even with the same target pose. Masked CFG 2.5 was best but under-opened the index (12.24 vs 17.25 degrees; 5.89 vs 9.34 in two cases). CFG 4.5 left remnants with little opening.
Full-noise SDXL / empty desk	Four nine-channel inpaint trials produced gray foreign material. Whole-hand and narrow-index empty-desk trials invented boxes or tools. Neither became the movement method.
Qwen Image Edit 2511	Transformer-only NF4: two attempts ran out of memory before denoising. Transformer + text-encoder NF4: two 40-step runs finished with noisy gray images / visible pose colors. Full BF16 was not tested.
ObjectClear / segmentation	20 ObjectClear calls: original masks left remnants; alternate seed helped one case; low CFG left edge remnants. Full-contour CFG 2.5 / seed 10031 was useful upstream. GrabCut confused shadows/gaps. Converted SAM-HQ produced stripes despite high predicted IoU; author SAM-HQ plus reviewed gap corrections was retained.
Desk / shadow variants	Offset/blended/revised texture donors left seams. Hard cutouts retained slivers or web notches. Continuous-root shadow fits made forearm patches; stationary-shadow inpaint retained ghosts; local shadow replacement made translucent brown web patches.

RECTANGULAR EDIT / input



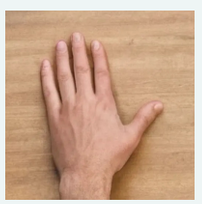
Archived actual experiment image.

RECTANGULAR EDIT / output



Archived actual experiment image.

QWEN NF4 / input



Archived actual experiment image.

QWEN NF4 / output



Archived actual experiment image.

Rectangular screening changed protected pixels in four of six outputs. The completed Qwen run shows gray noise and pose-guide color; completion did not make it a usable stimulus.

Earlier methods / geometry and source corrections

Retained upstream components are named explicitly; unused exploratory runners are absent from the clean production repository.

Raster / rigid-layer variants	Opaque whole-contour rasterization made a skin triangle; half-opacity topology hit fold guards; whole-mesh vector roots added jagged skin. Rigid upper-finger + local-root variants retained a palm flap; removing it left a one-pixel crack. Opaque index + harmonic piecewise-affine root was a useful five-case prototype, not a bulk acceptance guarantee.
Warp families / fresh pilots	Polar, continuous-root, sparse v1-v5, biharmonic v6/v7, harmonic v8, fair v9 and separator diagnostics were tried. Some reached 20/20 numerical passes while ownership, flaps, seams and realism still needed review. Fresh pilot v2 passed 14/20; v3/v4 passed 20/20 numerically. Visual problems remained.
V5 / 120 recorded calls	Shared nails/hand repairs and local movement repair superseded the first rough-knuckle cohort. Revised masks and seed-123 texture screening helped; reference 19 used seed 73. Garment trials included source-texture initialization; some invented trousers or limbs. Four reference-18 contact repairs left seams, causing temporary quarantine. V5 was superseded.
V6 SDXL / 42 trials	Local web/garment, whole-scene contact, detailed local garment, fabric-only and warped-fabric repairs were screened. Whole-scene repairs spoiled contact; local/fabric variants blurred anatomy or flattened clothing. All 14 warped-fabric candidates were user-rejected. One local reference-2 result was an intermediate selection.
V6 joint editor / 12 calls	Joint torso and both-sleeve edits from the built-in editor were selected upstream. Seeds/model internals were unavailable. Their results survive as frozen source pixels; this is not a reproducible seeded operation in the clean repository.
V7 / reference 2	Three geometry trials preceded one selected rim/root editor repair, retaining 6,088 original shaft pixels and restoring color after registration. All 20 V7 pairs were user-accepted, including reference 18. Four proposed SDXL repairs were not run because the GPU was busy. Detector-angle agreement was diagnostic.
Final choice	Freeze the accepted V7 pairs. Generate only guarded shared desk variants, then deterministic shared white balance and exact context exports. Hand segmentation, geometry and clothing edits are not rerun for each appearance family.

Exact negative prompts and pinned components

All current expansion values are centralized in config/recipe.json. The historical source and repair prompts are shown on their workflow pages.

Source appearance / negative	extra fingers, fused fingers, missing fingers, duplicate hand, distorted joints, transparent skin, plastic skin, halo, diagram, text, watermark
Shared hand / negative	cutout, halo, floating hand, sharp seam, webbed fingers, fused fingers, extra fingers, distorted nails, plastic skin, illustration, text, watermark
Movement repair / negative	scar, wound, rash, eczema, lesion, rough skin, skin disease, crust, burn, cutout, halo, floating hand, seam, webbed fingers, fused fingers, extra fingers, distorted nails, plastic skin, illustration, text, watermark
Desk expansion / negative	objects, clutter, extra hand, fingers, face, body, fabric, dark shadow, glossy reflection, cutout, floating, halo, hard seam, illustration, text, watermark
RealVisXL V4.0	SG161222/RealVisXL_V4.0 Revision: 26dfe44930964cd70d0a817b6d1cc945c130e38d
Canny ControlNet	xinsir/controlnet-canny-sdxl-1.0 Revision: 1271357eda52d54b857c650cacb5b51144643ccb
Fixed fp16 SDXL VAE	madebyollin/sdxl-vae-fp16-fix Revision: 207b116dae70ace3637169f1ddd2434b91b3a8cd
Shared sampling details	Both SDXL encoders receive identical positive and negative prompts; each must fit its 77-token limit. CPU-seeded generator; fp16; DPM++ multistep with Karras sigmas. Source creation uses text-to-image with Canny. Hand/movement/desk repairs use inpaint with a permission mask and protected native composition.
Exact reproducibility limits	Model revisions and package versions are pinned, but output also depends on GPU/runtime. Resume checks the recorded environment and GPU. Built-in editor source repairs expose neither a seed nor a pinned model revision; preserve their frozen result pixels.

Where we are, and what comes next

The accepted image batch is ready for a frozen analysis protocol. Scientific evaluation remains a separate phase.

Done / research and stimuli	Model/control/repair comparisons established the image-building approach. All 20 V7 source pairs were approved. Original production expanded them to 480 families / 2,880 PNGs; the user accepted the batch on 11 October with a reference-17 clothing flag.
Done / clean local project	Code is grouped into scripts/, tests/ and config/. README maps every file. Frozen inputs, original accepted dataset, its original mask provenance and the large interactive review stay local. The public repository carries compact approval metadata, this illustrated guide and the validation report. All 2,880 original image hashes and historical pixel checks pass. All 480 families differ inside the corrected guard; the validation report quantifies this separately. No accepted pixels are silently changed.
Next 1 / fresh CUDA canaries	Create the pinned Linux CUDA environment, download the three pinned models, run preflight and ten canaries, inspect all six conditions and record approval before a new full batch. CPU and exact composition checks pass; fresh GPU sampling of the clean code has not been exercised.
Next 2 / freeze analysis cohort	Choose the historical accepted batch or a newly accepted run explicitly. Audit upstream common origins before splitting; keep related sources and all appearance/context variants together. Repeat results without reference 17 (456 families / 2,736 PNGs). Define A-to-A and A-to-B timing before feature extraction.
Next 3 / visual representations	Compare CLIP/DINO with hand-only and simple low-level baselines using grouped held-out sources. Predefine the body-context x movement contrast, report uncertainty by source group, and inspect desk, white-balance and clothing confounds.
Next 4 / motor-response target	Confirm the intended human motor-evoked-potential protocol. The older report mentions thumb motion, whereas these stimuli use index movement. Establish a minimal spiking-network baseline and test the intended interaction on held-out source groups. Image acceptance alone is not physiological replication.

Evidence scope: saved source-generation and repair records; frozen V7 manifest and final acceptance; production recipe and accepted image hashes; archived trial ledgers and existing failure images; earlier TRDP-I report. Pinned model revisions and exact current prompts are in config/recipe.json. Earlier 252-family results are separate from this 480-family snapshot.